

# RTSO-2002

## Product Manual

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## Revision History

| Revision | Date       | Reason for change                                                                                                                                                                              | Applicable hardware version | Editor |
|----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------|
| V1.0     | 2022-11    | Initial release                                                                                                                                                                                | v1.0                        | RT0086 |
| V2.0     | 2023-05    | 1. The SIM card is updated from two to one, and the MiniPCle port and M.2B KEY port are shared<br>2. Replace the Sim card port model<br>3. Replace the model of the external output power port | v2.0                        | RT0086 |
| V2.1     | 2024-08-01 | Add RTST-2002-GMSL camera extension board and RTST-2002-SATA hard disk extension board information description                                                                                 | v2.0                        | RT0086 |
| V2.2     | 2026-03-20 | Modify the GPIO pin number                                                                                                                                                                     | v2.0                        | RT0086 |



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Electronic components and circuits are very sensitive to electrostatic discharge. Although our company designs anti-static protection for the main interfaces on the card when designing circuit board products, it is difficult to achieve anti-static safety protection for all components and circuits. Therefore, it is recommended to observe anti-static safety precautions when handling any circuit board component (including RTS0-2002). Anti-static safety protection measures include, but are not limited to the following:

- a) The smart box should be placed in an anti-static bag during transportation and storage, and then the board should not be taken out during installation and deployment.
- b) Before touching the smart box, discharge the static electricity stored in the body: wear a discharge grounding wrist strap.
- c) Operate the smart box only within the safe area of the electrostatic discharge point.
- d) Avoid moving smart boxes in carpeted areas.

## Precautions and after-sales maintenance

matters needing attention

Before using the product, please read this manual carefully and keep it for future reference;

- Please pay attention to and follow all warning and guidance information marked on the product;
- Please use matching power adapter to ensure the stability of voltage and current;
- Please use this product in a cool, dry and clean place;
- Do not use this product in cold and hot alternate environment to avoid condensation damage components;
- Do not splash any liquid on the product. Do not use organic solvent or corrosive liquid to clean the product;
- Do not use the product in dusty and messy environment. If it is not used for a long time, please pack the product;
- Do not use in the environment with excessive vibration, any dropping or knocking may damage the circuit and components;
- Do not plug and unplug the core board and peripheral modules when power is on;
- Please do not repair or disassemble the product by yourself. In case of any fault, please contact our company in time for maintenance;
- Do not modify or use unauthorized accessories by yourself, and the damage caused will not be warranted;



After sales maintenance

1) Warranty period

- Base plate, core plate : 3 year (non-human damage)
- Other peripherals sold by the company: 1 year (non-human damage)

2) Warranty description

- Within 7 days: the product (base plate, core module) is not damaged by human, our company will replace / repair it free of charge, and bear the return freight; (because the core module needs NVIDIA to confirm that it can meet the requirements of repair, it will take a long time, we will coordinate as soon as possible, please forgive for the inconvenience)
- From 7 days to 36 months: the product (base plate, core module) is not damaged by human, our company will repair it free of charge, and bear the return freight; (because the core module needs NVIDIA to confirm that it can meet the requirements of repair, it will take a long time, we will coordinate as soon as possible, please forgive for the inconvenience)
- Artificial damage in more than 3 year or 3 year: the product (carrier plate) shall be tested after it is sent to the customer, and the customer shall be informed of whether it can be repaired and the maintenance cost in detail. After reaching an agreement, the product shall be repaired and returned to the customer, and the company shall bear the return freight;
- The starting time shall be subject to the date of express delivery receipt;

3) Contact information

Official website: [www.realtimesai.com](http://www.realtimesai.com)

Taobao website: <https://shop340963258.taobao.com/>

Address: 11, block B, Heping Xiyuan, Heping West Street, Chaoyang District, Beijing

Attention: RMA

Tel: 400-100-8358

Mailing notice: contact with the company's sales department in advance, arrange technical support personnel to check and eliminate errors caused by misoperation as soon as possible, fill in the product after-sale return to factory maintenance form after verification, and send it to [rma@realtimes.cn](mailto:rma@realtimes.cn) Mail box, please attach the list of items to facilitate verification, so as to avoid loss and loss in the process of express delivery. The company does not receive any delivery

## Technical support and development customization

1. Scope of technical support

- 1) The company releases the electrical characteristics and use of industrial carrier boards and modules;
- 2) Physical dimension of hardware, relevant structure diagram and line sequence definition of specific interface;
- 3) Burn in verification of all BSP support packages provided by the company;
- 4) The company released burn environment construction, entry-level use. ;
- 5) Various peripheral module drivers released by the company;
- 6) The company's product fault diagnosis and after-sales maintenance services;

2. Scope of technical discussion

Due to the wide range of embedded system knowledge and various types of involvement, we can not guarantee that all kinds of questions can be answered one by one. The following content is not available for technical support, only suggestions can be provided.

- 1) Knowledge beyond the course published by our company;



- 2) Specific software program design;
- 3) Technical support for industrial carrier not issued by the company;
- 4) All kinds of driving support for industrial carrier board not issued by the company;
- 5) Hardware principle and drive design of peripheral module not issued by our company;

### 3. Technical support mode

- 1) Official website or email questions (recommended): <https://www.realtimesai.com/cn/download.html>  
[techsupport@realtimes.cn](mailto:techsupport@realtimes.cn)
- 2) Official Taobao through Alibaba Wangwang consultation: <https://shop340963258.taobao.com/>
- 3) Wechat group consultation (wechat Group No. consults Taobao customer service or sales, and Taobao purchase order No. needs to be provided for verification);
- 4) Technical support email: [techsupport@realtimes.cn](mailto:techsupport@realtimes.cn)
- 5) Tel: 400-100-8358

### 4. Technical support time

Monday to Friday; 8:30-12:00 am; 1:00-17:30 PM;

The company arranges the rest according to the national legal holidays, during which it may not be able to provide technical support, please send the problem to the technical support email. We will reply to you as soon as possible on weekdays.

### 5. Complaints and suggestions

If you are not satisfied with us or have suggestions, you can send an email to [yu.qin@realtimes.cn](mailto:yu.qin@realtimes.cn) For feedback, please call 400-100-8358 for further improvement.

### 6. Customized development services

The company provides the embedded operating system driver based on NVIDIA Jetson series and the paid customized development service of hardware carrier board to shorten your product development cycle.

Please email the request to [info@realtimes.cn](mailto:info@realtimes.cn)

## Data acquisition and subsequent update

### 1. Access to information

#### Download on our website

The company's website contains supporting information of its products, including product user manual, NVIDIA Jetson series module data manual, BSP driver support package for carrier board, supporting peripheral driver files, interface test verification method, FAQ, system burning guide, etc. get into [www.realtimesai.com](http://www.realtimesai.com), select "data download" in the navigation bar, find the data you need, and click download.

### 2. Subsequent updates

Updates of subsequent documents, BSP, driver files and other official account will be updated in time. We will pay close attention to our developments in order to ensure that your information is up to date. We will push through WeChat public.

# Product presentation

## 1 Introduction

RTSO-2002 project is based on NVIDIA Jetson AGX Orin's powerful computing power, video engine and rich interface resources, designed and developed for specific target customers with strong environmental awareness capabilities of visual products. Use the various interface resources of the module to expand 5 independent Gigabit Ethernet, 4 USB3.0 interfaces and other conventional communication interfaces, and reserve high-speed connectors for later expansion of camera, SATA and other functions. The product has a wide range of applications in the field of pan-automatic driving, AGV car, logistics vehicle, robot and so on.

### 1.1 Features

| Interface                | PN | RTSO-2002                                                                                                                                                                                 |
|--------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI Core                  |    | AGX Orin                                                                                                                                                                                  |
| Preinstalled system      |    | Ubuntu20.04                                                                                                                                                                               |
| Display                  |    | 1 x HDMI TYPE A interface;                                                                                                                                                                |
| USB                      |    | 4 x USB TYPE A 3.0 ports GEN1, support usb2.0, usb3.1 signals, provide 1A output current; 1 x Micro USB port for system flash;                                                            |
| Network                  |    | 5 x Gigabit Ethernet (10/100/1000Mbps adaptive; Half-duplex/full-duplex adaptive)                                                                                                         |
| Storage                  |    | 1 x Micro SD card slot;                                                                                                                                                                   |
| SIM                      |    | 1 x Micro SIM card slot;                                                                                                                                                                  |
| PCIe                     |    | 1 x Mini PCIe port,, support 4G/WIFI5/WIFI6;                                                                                                                                              |
| M.2                      |    | 1 x M.2 key M port,Support NVMe SSD hard disk; 1 x M.2 key B port,Support 4G/5G;                                                                                                          |
| Output Power             |    | 1 x Power output port (consistent with input voltage), the maximum bearing current 6A;                                                                                                    |
| Expansion Interface      |    | 1 x High-speed expansion connector, 2 x Low-speed expansion double-row pins; (Scalable GMSL camera or SATA storage)                                                                       |
| Multi-function interface |    | 2 x Multifunctional double-row pins; (support 2 x I2C, 2 x SPI, 1 x Debug RS232, 2 x RS232, 2 x RS485, 4 x GPI, 4 x GPO, 2 x CAN, 1 x SYNC_IN, 1 x SYNC_OUT, 1 x SYNC_PPS, 1 x POWER OUT) |
| RTC                      |    | 2 x RTC battery ports, 1 x CR2032 button battery support, 1 x 2Pin socket;                                                                                                                |
| FAN                      |    | Support 5V/12V fan (optional) (default 5V, if you need 12V fan, please contact to confirm before ordering);                                                                               |
| Audio                    |    | 1 x Audio port, support Line_IN, MIC_IN, Line_OUT;                                                                                                                                        |
| Encryption Chip          |    | It supports encryption function and meets the design requirements of EAL4+ security level. Part number LKT4110U;                                                                          |
| Indicator Light          |    | 4 x External indicators, including POWER, USER, ALARM, HDD;                                                                                                                               |
| Button                   |    | 1 x Reset button; 1 x Recovery button; 1 x Power button;                                                                                                                                  |
| Size                     |    | 200mm×200mm×28mm                                                                                                                                                                          |
| Power Requirement        |    | 9-36VDC input voltage range                                                                                                                                                               |



|                       |            |
|-----------------------|------------|
| Operating Temperature | -20℃~+80℃  |
| Weight                | About 395g |

## 1.2 Ordering Information

| Ordering Information    | Ordering Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RTSO-2002               | Supports NVIDIA Jetson AGX Orin modules, 1 x HDMI 2.0, 4 x USB 3.0, 5 x GbE, 1 x Debug RS232, 2 x RS232, 2 x RS485, 4 x GPI, 4 x GPO, 1 x MiniPCie, 1 x M.2 B Key, 1 x M.2 M Key, 2 x I2C, 2 x SPI, 1 x Audio, 1 x SYNC_IN, 1 x SYNC_OUT, 4 x GPI, 4 x GPO, 1 x minipcie, 1 x M.2 M key, 2 x I2C, 2 x SPI, 1 x Audio, 2 x CAN, 1 x sync_in, 1 x sync_out, 1 x SYNC_PPS, 1 x POWER OUT, 1 x FAN, 2 x RTC, 3C certification, provides the Realtimes_L4T_R35.1.0_rtso-2002-v2.0_orin.tbz2 software support package. |
| RTST-2002-GMSL (Option) | GMSL camera extension board, adapted to RTSO-2002 carrier board, supports 8-channel GMSL2 camera                                                                                                                                                                                                                                                                                                                                                                                                                 |
| RTST-2002-SATA (Option) | SATA expansion board, suitable for RTSO-2002 carrier board, supports 2 channels of SATA3.0 interface (can connect 3.5 inch and 2.5 inch SATA disk)                                                                                                                                                                                                                                                                                                                                                               |

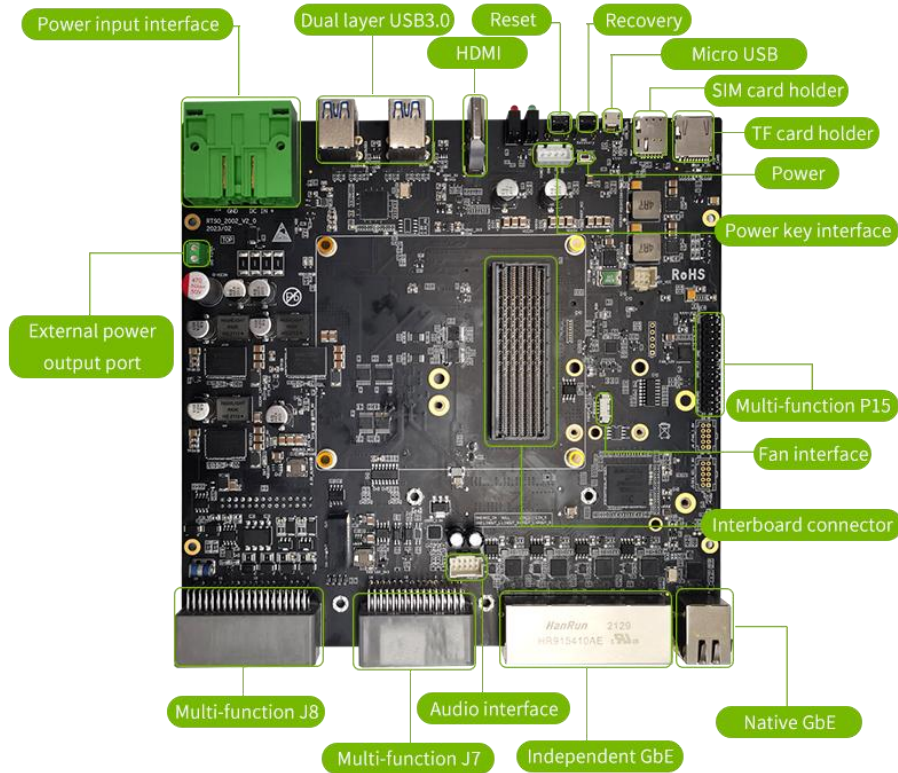
## 1.3 Adaptive cable package configuration

RTSO-2002 supporting cable package ordering model: RTSO-2002\_Cables. Contains the following components:

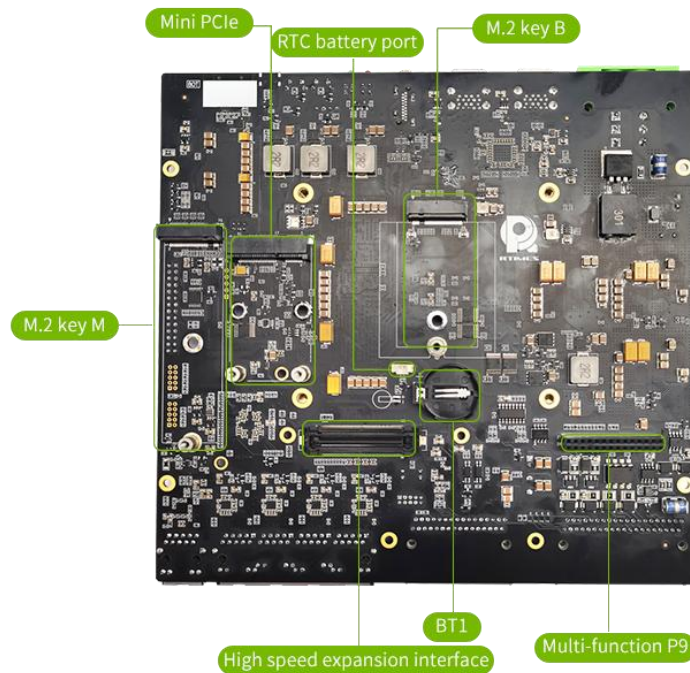
| Ordering Information | Quantity | Functional description |
|----------------------|----------|------------------------|
| RTSO-2002_Cables     | 1        | Matching bag           |

# Product specifications

## 1 Connector Locations

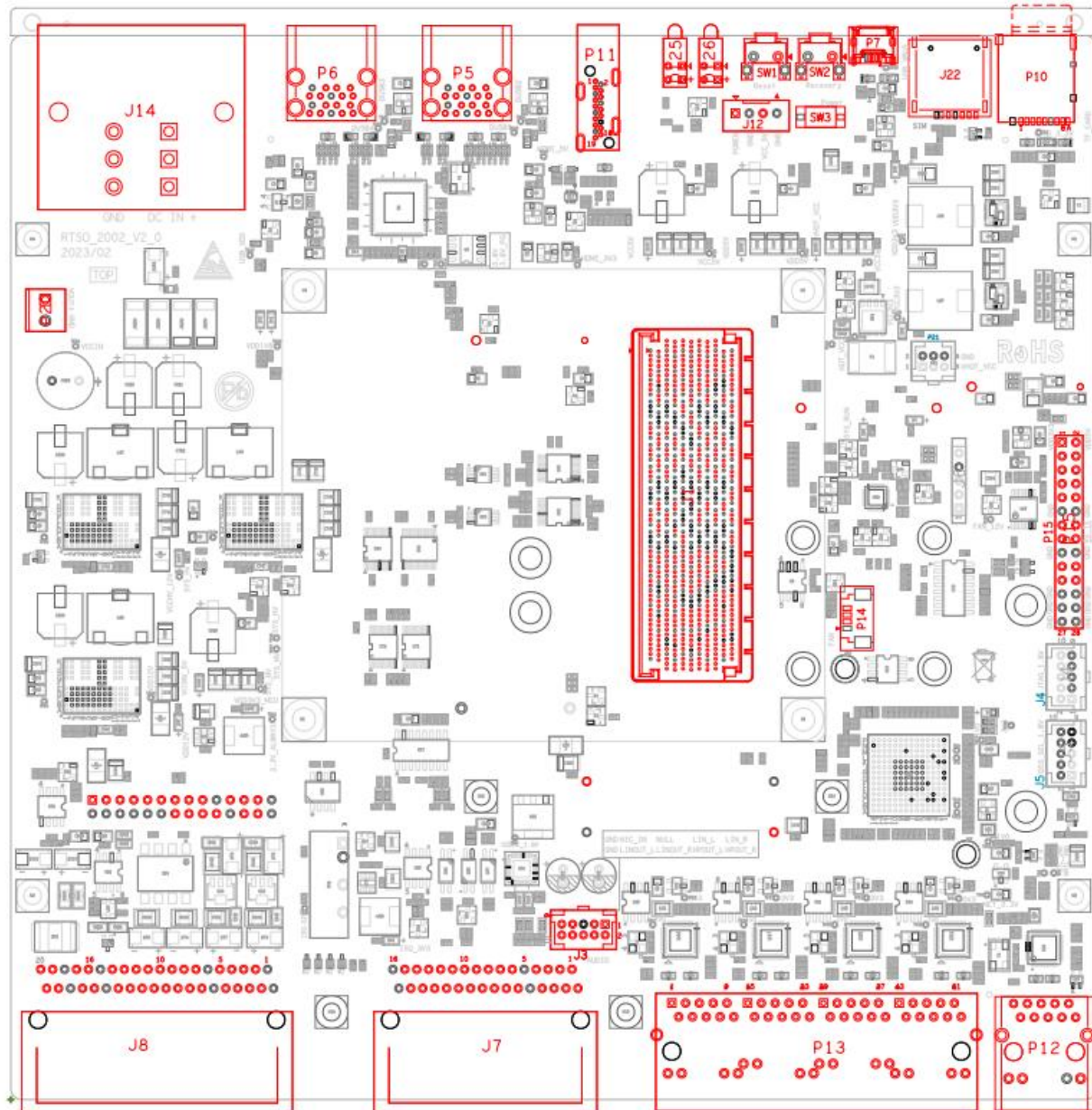


RTSO-2002 FRONT

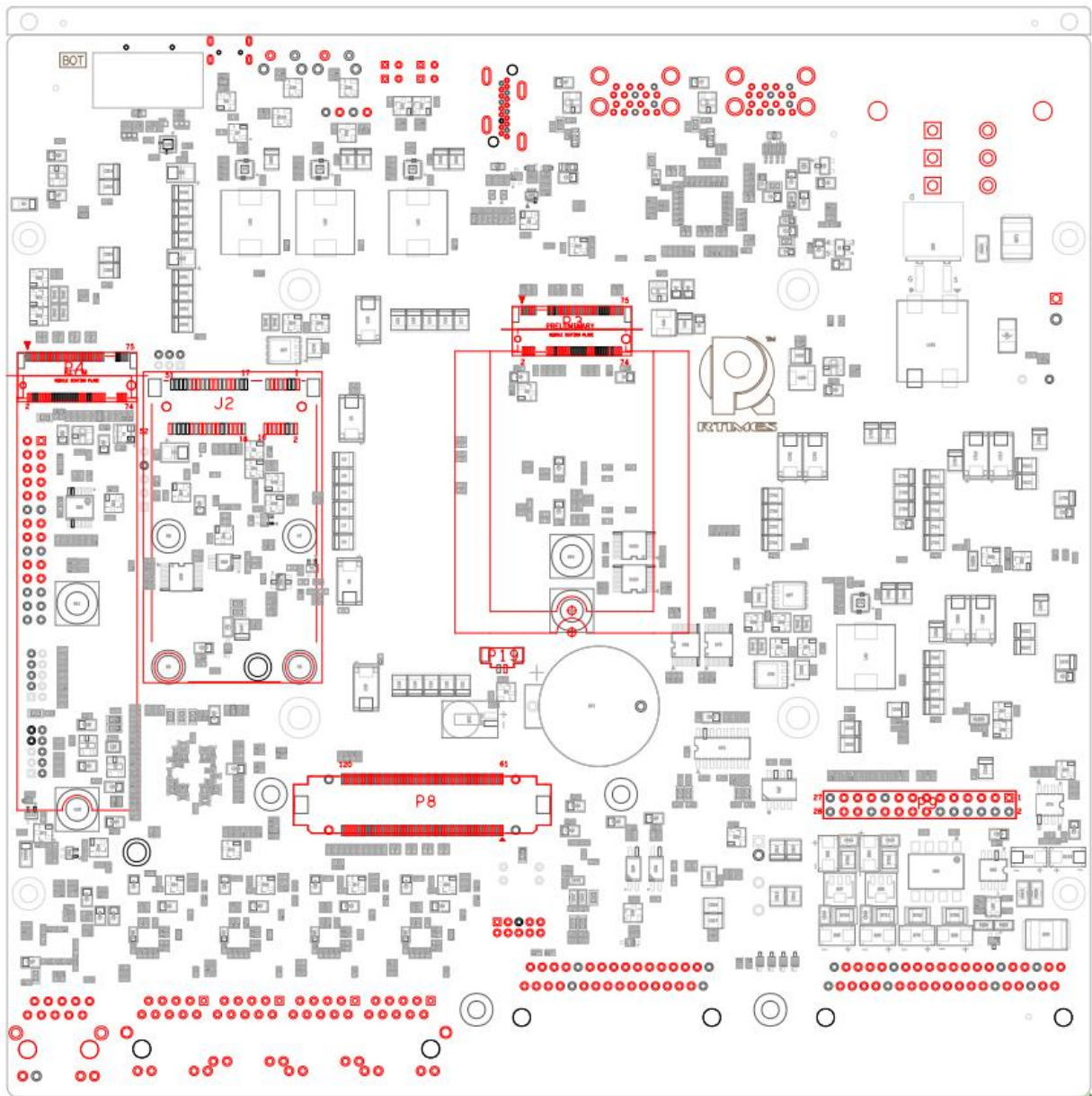


RTSO-2002 BACK

## 1.1 Component location diagram



RTSO-2002 FRONT



RTSO-2002 BACK

## 1.2 Functional connector

| Marking | Function description                                            |
|---------|-----------------------------------------------------------------|
| J1      | 699-Pin that connects to the NVIDIA Jetson AGX Orin core module |
| P12     | Native Gigabit interface                                        |
| P13     | Four independent Gigabit network ports                          |
| J22     | Self-ejecting SIM card with MiniPCIe port and M.2B KEY port     |
| P5/P6   | Dual layer USB3.0 interface                                     |
| P7      | MicroUSB interface (system flash port)                          |
| P11     | HDMI interface                                                  |
| J2      | MiniPCIe interface                                              |
| P3      | M.2 KEY B interface                                             |
| P4      | M.2 KEY M interface                                             |
| P10     | TF card interface                                               |
| J3      | Audio interface                                                 |
| P14     | Fan interface                                                   |
| P15/P9  | Multifunction extension interface                               |
| J12     | Power button interface                                          |
| J7/J8   | External multifunctional interface                              |
| P8      | High speed expansion interface                                  |
| BT1/P19 | RTC battery port                                                |
| P20     | External power output port                                      |
| J14     | Power input interface                                           |

## 1.3 Button

| Marking | Function description                                               |
|---------|--------------------------------------------------------------------|
| SW1     | RESET button,Indicates the system reset button                     |
| SW2     | RECOVERY button, enable the core module to enter the recovery mode |
| SW3     | POWER button, Switches on and off the machine                      |

## 1.4 LED indicator light

| Marking           | Function description                                                                   |
|-------------------|----------------------------------------------------------------------------------------|
| D125-substratum   | HDD LED: Red light, SSD running indicator, off when the storage device is not inserted |
| D125-superstratum | ALEAM LED: Red light, off by default, software programmable. (GPIO Number 486)         |
| D126-substratum   | USER LED: Green light, off by default, software programmable. (GPIO No. 488)           |
| D126-superstratum | POWER LED: Green light, power indicator, blinking during normal operation              |



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## Note:

### USER LED:

```
sudo echo 488 > /sys/class/gpio/export
echo 1 or 0 > /sys/class/gpio/PAC.02/value 1: 0n 0:0ff
```

### ALEAM LED:

```
sudo echo 486 > /sys/class/gpio/export
echo 1 or 0 > /sys/class/gpio/PAC.00/value 1: 0n 0:0ff
```

## 1.5 Encryption chip

|                               |                                                                         |
|-------------------------------|-------------------------------------------------------------------------|
| <b>Function</b>               | Key code protection, contrast authentication, parameter protection, etc |
| <b>Chip type</b>              | LKT4110U                                                                |
| <b>Chip parameter</b>         | 28K-Bytes program storage, 4K-Bytes NVM data storage, 4K-Bytes RAM      |
| <b>Main machine interface</b> | /dev/I2C-2                                                              |
| <b>Chip address</b>           | 0x50                                                                    |

Note: \* Please refer to the LKT4110U chip development manual for detailed usage

## 2 External ports on the GMSL2 expansion board

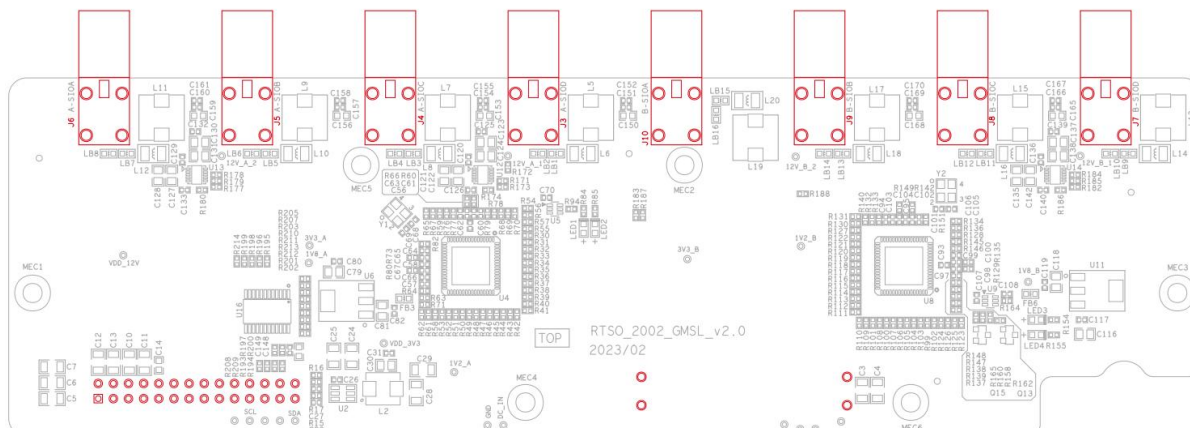


GMSL Expansion Board Front

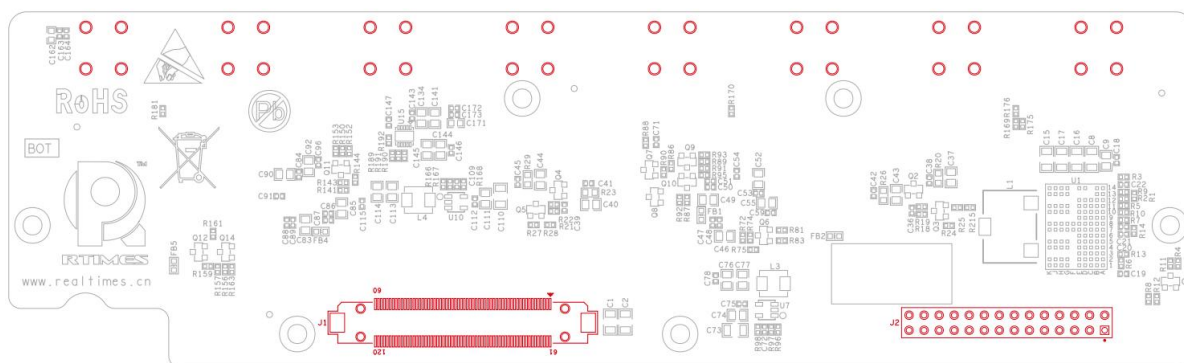


GMSL Expansion Board Back

## 2.1 Component location diagram



GMSL Expansion Board Front

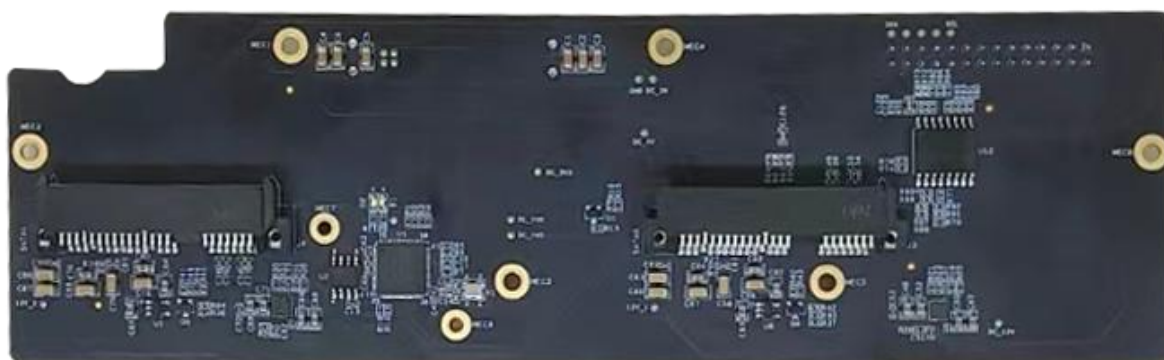


GMSL Expansion Board Back

## 2.2 Functional connector

| Marking                  | Function description                                                      |
|--------------------------|---------------------------------------------------------------------------|
| J1                       | High speed signal connector for connecting carrier board                  |
| J2                       | Low speed signal connector for connecting carrier board                   |
| J6,J5,J4,J3,J10,J9,J8,J7 | GSML2 camera interface,From left to right, the device nodes are: video0-7 |

## 3 External ports on the SATA expansion board

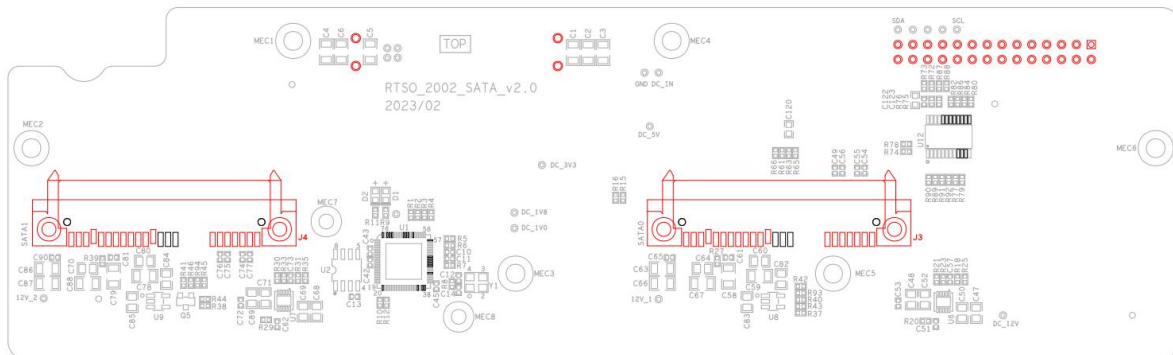


SATA Expansion Board Front

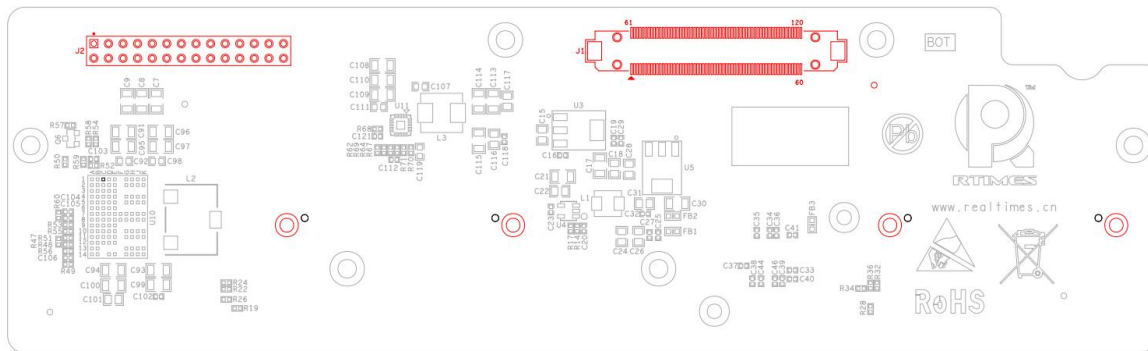


SATA Expansion Board Back

### 3.1 Component location diagram



SATA Expansion Board Front



SATA Expansion Board Back

### 3.2 Functional connector

| Marking | Function description                                     |
|---------|----------------------------------------------------------|
| J1      | High speed signal connector for connecting carrier board |
| J2      | Low speed signal connector for connecting carrier board  |
| J3、J4   | 22Pin SATA hard drive socket                             |

## 4 Connectors Description

### 4.1 Connectors Description

|                   |                                                                                                                                      |                                                                                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Function</b>   | Connect the NVIDIA Jetson AGX Orin core module                                                                                       |  |
| <b>Marking</b>    | J1                                                                                                                                   |                                                                                     |
| <b>Type</b>       | 699-Pin SO-DIMM                                                                                                                      |                                                                                     |
| <b>Pin define</b> | For pin definitions of this connector, refer to the pin definition instructions in the NVIDIA Jetson AGX Orin Core Module data Book. |                                                                                     |

### 4.2 Gigabit network interface

|            |                           |        |     |        |
|------------|---------------------------|--------|-----|--------|
| Function   | Gigabit network connector |        |     |        |
| Marking    | P12                       |        |     |        |
| Type       | RJ45/48F-01GY2DPL2NL      |        |     |        |
| Pin define |                           |        |     |        |
|            | Pin                       | Signal | Pin | Signal |
|            | 1                         | TP0+   | 2   | TP0-   |
|            | 3                         | TP1+   | 4   | TP1-   |
|            | 5                         | TP2+   | 6   | TP2-   |
|            | 7                         | TP3+   | 8   | TP3-   |

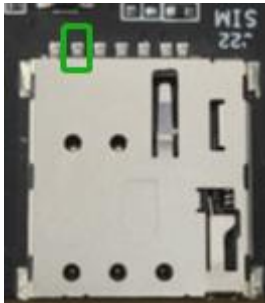


### 4.3 Four Gigabit network ports

|                                                                                        |                           |        |     |        |
|----------------------------------------------------------------------------------------|---------------------------|--------|-----|--------|
| Function                                                                               | Gigabit network connector |        |     |        |
| Marking                                                                                | P13                       |        |     |        |
| Type                                                                                   | RJ45/48F-01GY2DPL2NL      |        |     |        |
| Pin define                                                                             |                           |        |     |        |
|                                                                                        | Pin                       | Signal | Pin | Signal |
|                                                                                        | 1                         | TP0+   | 2   | TP0-   |
|                                                                                        | 3                         | TP1+   | 4   | TP1-   |
|                                                                                        | 5                         | TP2+   | 6   | TP2-   |
|                                                                                        | 7                         | TP3+   | 8   | TP3-   |
| Note: The four network ports from left to right correspond to eth1-eth4 of the system. |                           |        |     |        |

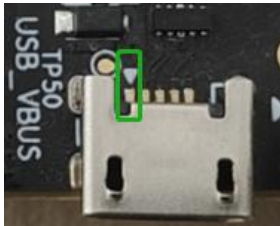


#### 4.4 SIM card slot

| Function                                                                                      | SIM card slot                                                                                                                                                                                                                                  |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|--------|-----|--------|-----|--------|---|-----|---|-----|---|-----|---|-----|---|-----|---|------|
| Marking                                                                                       | J22                                                                                                                                                                                                                                            |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
| Type                                                                                          | SIM card slot/SNO-1303-X                                                                                                                                                                                                                       |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
| Pin define                                                                                    | <table><tr><th>Pin</th><th>Signal</th><th>Pin</th><th>Signal</th></tr><tr><td>1</td><td>VCC</td><td>2</td><td>RST</td></tr><tr><td>3</td><td>CLK</td><td>4</td><td>GND</td></tr><tr><td>5</td><td>VPP</td><td>6</td><td>DATA</td></tr></table> |        |     |        | Pin | Signal | Pin | Signal | 1 | VCC | 2 | RST | 3 | CLK | 4 | GND | 5 | VPP | 6 | DATA |
|                                                                                               | Pin                                                                                                                                                                                                                                            | Signal | Pin | Signal |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|                                                                                               | 1                                                                                                                                                                                                                                              | VCC    | 2   | RST    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|                                                                                               | 3                                                                                                                                                                                                                                              | CLK    | 4   | GND    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|                                                                                               | 5                                                                                                                                                                                                                                              | VPP    | 6   | DATA   |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|                                                                                               | Pin1: marked in the green box on the right.                                                                                                                                                                                                    |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
| Note: SIM card gap facing inward, 4G, 5G network switching, see Appendix I below the document |                                                                                                                                                                                                                                                |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |
|            |                                                                                                                                                                                                                                                |        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |      |

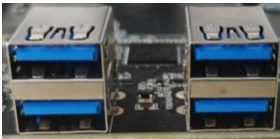
#### 4.5 Micro USB

|            |                                             |        |     |        |
|------------|---------------------------------------------|--------|-----|--------|
| Function   | Micro USB                                   |        |     |        |
| Marking    | P7                                          |        |     |        |
| Type       | Micro USB/MC-142PD                          |        |     |        |
| Pin define |                                             |        |     |        |
|            | Pin                                         | Signal | Pin | Signal |
|            | 1                                           | VBUS   | 2   | DN     |
|            | 3                                           | DP     | 4   | ID     |
|            | 5                                           | GND    |     |        |
|            | Pin1: marked in the green box on the right. |        |     |        |

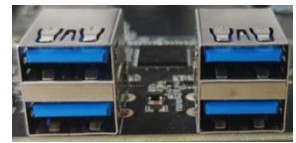


## 4.6 USB3.0 interface

|            |                                      |              |     |            |
|------------|--------------------------------------|--------------|-----|------------|
| Function   | Dual layer USB3.0 connector          |              |     |            |
| Marking    | P5/P6                                |              |     |            |
| Type       | Dual-layer USB TYPE-A port /U30-021D |              |     |            |
| Pin define | Level                                | Superstratum |     |            |
|            | Type                                 | USB 3.0      |     |            |
|            | Pin                                  | Signal       | Pin | Signal     |
|            | 10                                   | VBUS         | 11  | USB 2.0 D- |
|            | 12                                   | USB 2.0 D+   | 13  | GND        |
|            | 14                                   | SSRX-        | 15  | SSRX+      |
|            | 16                                   | GND          | 17  | SSTX-      |
|            | 18                                   | SSTX+        |     |            |
|            | Level                                | Substratum   |     |            |
|            | Level                                | USB 3.0      |     |            |
|            | Pin                                  | Signal       | Pin | Signal     |
|            | 1                                    | VBUS         | 2   | USB 2.0 D- |
|            | 3                                    | USB 2.0 D+   | 4   | GND        |
|            | 5                                    | SSRX-        | 6   | SSRX+      |
|            | 7                                    | GND          | 8   | SSTX-      |
|            | 9                                    | SSTX+        |     |            |



Note: Four USB3.0 TYPE A ports share 5Gbps bandwidth.



## 4.7 Display interface

|            |                                    |                 |        |                |
|------------|------------------------------------|-----------------|--------|----------------|
| Function   | HDMI Display interface             |                 |        |                |
| Marking    | P11                                |                 |        |                |
| Type       | Vertical HDMI interface /HDMI 019F |                 |        |                |
| Pin define | Pin                                |                 | Signal |                |
|            | 1                                  | TMDS Data2+     | 2      | TMDS Data2 GND |
|            | 3                                  | TMDS Data2-     | 4      | TMDS Data1+    |
|            | 5                                  | TMDS Data1 GND  | 6      | TMDS Data1-    |
|            | 7                                  | TMDS Data0+     | 8      | TMDS Data0 GND |
|            | 9                                  | TMDS Data0-     | 10     | TMDS Clock+    |
|            | 11                                 | TMDS Clock GND  | 12     | TMDS Clock-    |
|            | 13                                 | CEC             | 14     | No Connect     |
|            | 15                                 | DDC clock       | 16     | DDC data       |
|            | 17                                 | DDC GND         | 18     | +5V Power      |
|            | 19                                 | Hot Plug Detect |        |                |





## 4.8 Extension interface 1

|            |                                   |               |     |                  |
|------------|-----------------------------------|---------------|-----|------------------|
| Function   | Multifunction extension interface |               |     |                  |
| Marking    | P15                               |               |     |                  |
| Type       | 2.54mm-2x14pin double row pins    |               |     |                  |
| Pin define | Pin                               | Signal        | Pin | Signal           |
|            | 1                                 | VCC-3V3       | 2   | VCC-5V           |
|            | 3                                 | SPI1_CLK_3V3  | 4   | SPI3_CLK_3V3     |
|            | 5                                 | SPI1_MOSI_3V3 | 6   | SPI3_MOSI_3V3    |
|            | 7                                 | SPI1_MISO_3V3 | 8   | SPI3_MISO_3V3    |
|            | 9                                 | SPI1_CS_3V3   | 10  | SPI3_CS_3V3      |
|            | 11                                | GND           | 12  | GND              |
|            | 13                                | I2C0_SCL_3V3  | 14  | TXD-232(CONSOLE) |
|            | 15                                | I2C0_SDA_3V3  | 16  | RXD-232(CONSOLE) |
|            | 17                                | GND           | 18  | GND              |
|            | 19                                | I2C1_SCL_3V3  | 20  | PWM01 3V3        |
|            | 21                                | I2C1_SDA_3V3  | 22  | GPIO29_3V3       |
|            | 23                                | GND           | 24  | GND              |
|            | 25                                | GND           | 26  | GND              |
|            | 27                                | GND           | 28  | GND              |

Pin1: green box marked on the right.

Pin2: red box marked on the right.

In Linux,The mapping file for SPI1 is spi0.0 in the /dev directory

In Linux,The mapping file for SPI3 is spi2.0 in the /dev directory

In Linux,The mapping file for I2C0 is I2C-16 in the /dev directory

In Linux,The mapping file for I2C1 is I2C-15 in the /dev directory

**Note:**

1.SPI, I2C, PWM01\_3V3, GPIO29\_3V3 are 3.3V level standards

2.TXD-232 and RXD-232 are debugging serial ports with standard RS-232 level. The serial port configuration parameters are: 115200, 8, n, 1

3.spi driver loading method :sudo modprobe spidev

4. For details about how to use GPIO01 and GPIO02, see Appendix 5 below the document

## 4.9 Extension interface 2

|                 |                                   |  |
|-----------------|-----------------------------------|--|
| <b>Function</b> | Multifunction extension interface |  |
| <b>Marking</b>  | P9                                |  |
| <b>Type</b>     | 2.54mm-2x14pin Dual female holder |  |

| Pin define | Pin                                                                                                      | Signal        | Pin | Signal      |
|------------|----------------------------------------------------------------------------------------------------------|---------------|-----|-------------|
|            | 1                                                                                                        | VCC_IN        | 2   | GND         |
|            | 3                                                                                                        | VCC_IN        | 4   | GND         |
|            | 5                                                                                                        | VCC_IN        | 6   | GND         |
|            | 7                                                                                                        | VCC_IN        | 8   | GND         |
|            | 9                                                                                                        | VCC_IN        | 10  | GND         |
|            | 11                                                                                                       | VCC_IN        | 12  | GND         |
|            | 13                                                                                                       | ORIN_GPIO0_IN | 14  | MCLK02-1V8  |
|            | 15                                                                                                       | ORIN_GPIO1_IN | 16  | MCLK03-1V8  |
|            | 17                                                                                                       | ORIN_GPIO2_IN | 18  | MCLK04-1V8  |
|            | 19                                                                                                       | GND           | 20  | MCLK05-1V8  |
|            | 21                                                                                                       | GPIO3C00      | 22  | GND         |
|            | 23                                                                                                       | GPIO3C01      | 24  | I2C_SCL_3V3 |
|            | 25                                                                                                       | GPIO3C02      | 26  | I2C_SDA_3V3 |
|            | 27                                                                                                       | GND           | 28  | GND         |
|            | Pin1: green box marked on the right.                                                                     |               |     |             |
|            | Pin2: red box marked on the right.                                                                       |               |     |             |
|            | VCC_IN:9~36V Consistent with the input voltage                                                           |               |     |             |
|            | In Linux, The mapping file for I2C is i2c-11 in the /dev directory                                       |               |     |             |
|            | <b>Note:</b>                                                                                             |               |     |             |
|            | 1, I2C, ORIN_GPIO0_IN, ORIN_GPIO1_IN, ORIN_GPIO2_IN are 3.3V level standards                             |               |     |             |
|            | 2, MCLK02-1V8, MCLK03-1V8, MCLK04-1V8, MCLK05-1V8, GPIO3C00, GPIO3C01, GPIO3C02 are 1.8V level standards |               |     |             |
|            | * Used with high-speed connector P8, cannot be used alone                                                |               |     |             |



#### 4.10 High speed expansion interface

| Function   | High speed expansion interface |             |     |             |
|------------|--------------------------------|-------------|-----|-------------|
| Marking    | P8                             |             |     |             |
| Type       | FX23L-120S-0.5SV               |             |     |             |
| Pin define | Pin                            | Signal      | Pin | Signal      |
|            | 123                            | GND         | 121 | VCC_IN      |
|            | 1                              | CSI_6_D0_P  | 61  | CSI_0_D0_P  |
|            | 2                              | CSI_6_D0_N  | 62  | CSI_0_D0_N  |
|            | 3                              | GND         | 63  | GND         |
|            | 4                              | CSI_6_D1_P  | 64  | CSI_0_D1_P  |
|            | 5                              | CSI_6_D1_N  | 65  | CSI_0_D1_N  |
|            | 6                              | GND         | 66  | GND         |
|            | 7                              | CSI_6_CLK_P | 67  | CSI_0_CLK_P |
|            | 8                              | CSI_6_CLK_N | 68  | CSI_0_CLK_N |





|    |              |             |            |             |
|----|--------------|-------------|------------|-------------|
|    | 9            | GND         | 69         | GND         |
|    | 10           | CSI_7_D0_P  | 70         | CSI_1_D0_P  |
|    | 11           | CSI_7_D0_N  | 71         | CSI_1_D0_N  |
|    | 12           | GND         | 72         | GND         |
|    | 13           | CSI_7_D1_P  | 73         | CSI_1_D1_P  |
|    | 14           | CSI_7_D1_N  | 74         | CSI_1_D1_N  |
|    | 15           | GND         | 75         | GND         |
|    | 16           | CSI_7_CLK_P | 76         | CSI_1_CLK_P |
|    | 17           | CSI_7_CLK_N | 77         | CSI_1_CLK_N |
|    | 18           | GND         | 78         | GND         |
|    | 19           | PCIE7_CLK_P | 79         | CSI_2_D0_P  |
|    | 20           | PCIE7_CLK_N | 80         | CSI_2_D0_N  |
|    | 21           | GND         | 81         | GND         |
|    | 22           | UPHY_TX2_P  | 82         | CSI_2_D1_P  |
|    | 23           | UPHY_TX2_N  | 83         | CSI_2_D1_N  |
|    | 24           | GND         | 84         | GND         |
|    | 25           | UPHY_RX2_P  | 85         | CSI_2_CLK_P |
|    | 26           | UPHY_RX2_N  | 86         | CSI_2_CLK_N |
|    | 27           | GND         | 87         | GND         |
|    | 28           | UPHY_TX3_P  | 88         | CSI_3_D0_P  |
|    | 29           | UPHY_TX3_N  | 89         | CSI_3_D0_N  |
|    | 30           | GND         | 90         | GND         |
|    | 31           | UPHY_RX3_P  | 91         | CSI_3_D1_P  |
|    | 32           | UPHY_RX3_N  | 92         | CSI_3_D1_N  |
|    | 33           | GND         | 93         | GND         |
|    | 34           | UPHY_TX4_P  | 94         | CSI_3_CLK_P |
|    | 35           | UPHY_TX4_N  | 95         | CSI_3_CLK_N |
|    | 36           | GND         | 96         | GND         |
|    | 37           | UPHY_RX4_P  | 97         | CSI_4_D0_P  |
|    | 38           | UPHY_RX4_N  | 98         | CSI_4_D0_N  |
|    | 39           | GND         | 99         | GND         |
|    | 40           | UPHY_TX5_P  | 100        | CSI_4_D1_P  |
|    | 41           | UPHY_TX5_N  | 101        | CSI_4_D1_N  |
|    | 42           | GND         | 102        | GND         |
|    | 43           | UPHY_RX4_P  | 103        | CSI_4_CLK_P |
|    | 44           | UPHY_RX4_N  | 104        | CSI_4_CLK_N |
|    | 45           | GND         | 105        | GND         |
|    | 46           | PCIE_WAKE   | 106        | CSI_5_D0_P  |
|    | 47           | PCIE7_RST   | 107        | CSI_5_D0_N  |
|    | 48           | GND         | 108        | GND         |
| 49 | PCIE7_CLKREQ | 109         | CSI_5_D1_P |             |
| 50 | GND          | 110         | CSI_5_D1_N |             |

|     |               |     |                |
|-----|---------------|-----|----------------|
| 51  | GND           | 111 | GND            |
| 52  | I2C2_CLK_1V8  | 112 | CSI_5_CLK_P    |
| 53  | I2C2_SDA_1V8  | 113 | CSI_5_CLK_N    |
| 54  | GND           | 114 | GND            |
| 55  | I2C8_CLK_1V8  | 115 | GPIO6_AON_3V3  |
| 56  | I2C8_SDA_1V8  | 116 | GPIO7_AON_3V3  |
| 57  | GPIO3_OUT_3V3 | 117 | GND            |
| 58  | GPIO4_OUT_3V3 | 118 | CARRIER_PWR_ON |
| 59  | GPIO3A06      | 119 | GPIO3A05       |
| 60  | GND           | 120 | GND            |
| 122 | VCC_IN        | 124 | GND            |

Pin1: green box marked on the right.  
Pin2: red box marked on the right.  
VCC\_IN:9~36V Consistent with the input voltage  
In Linux,The mapping file for I2C2 is I2C-1 in the /dev directory  
In Linux,The mapping file for I2C8 is I2C-7 in the /dev directory  
**Note: Used with SATA, GMSL expansion board, can not be used alone.**

#### 4.11 Multifunctional connector 1

|            |                                       |                  |     |               |
|------------|---------------------------------------|------------------|-----|---------------|
| Function   | Multifunctional connector             |                  |     |               |
| Marking    | J7                                    |                  |     |               |
| Type       | Multifunctional connector/ MX34032NF2 |                  |     |               |
| Pin define | Pin                                   | Signal           | Pin | Signal        |
|            | 1                                     | FAN_VCC          | 17  | SPI3_CLK_3V3  |
|            | 2                                     | FAN_VCC          | 18  | SPI3_MOSI_3V3 |
|            | 3                                     | FAN_PWM_OUT      | 19  | SPI3_MISO_3V3 |
|            | 4                                     | FAN_TACH_IN      | 20  | SPI3_CS_3V3   |
|            | 5                                     | GND              | 21  | GND           |
|            | 6                                     | MIC_IN           | 22  | LINEOUT       |
|            | 7                                     | LINELIN          | 23  | LINEROUT      |
|            | 8                                     | LINERIN          | 24  | AUDIO_GND     |
|            | 9                                     | AUDIO_GND        | 25  | AUDIO_GND     |
|            | 10                                    | ISO_SYNC_IN      | 26  | ISO_SYNC_IN_B |
|            | 11                                    | ISO_PPS_IN_3V3   | 27  | ISO_SYNC_IN_A |
|            | 12                                    | ISO_SYNC_OUT_3V3 | 28  | ISO_GND       |
|            | 13                                    | ISO_GND          | 29  | ISO_GND       |
|            | 14                                    | I2C1_SCL_3V3     | 30  | I2C0_SCL_3V3  |
|            | 15                                    | I2C1_SDA_3V3     | 31  | I2C0_SDA_3V3  |
|            | 16                                    | GND              | 32  | GND           |





|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>Pin1: green box marked on the right.</p> <p>Pin2: red box marked on the right.</p> <p>FAN_VCC:5V or 12V</p> <p>In Linux,the mapping file for SPI3 is spi2.0 in the /dev directory</p> <p>In Linux,the mapping file for I2C0 is I2C-16 in the /dev directory</p> <p>In Linux,the mapping file for I2C1 is I2C-15 in the /dev directory</p> <p><b>Note:</b></p> <p>1, SPI, I2C, ISO_SYNC_OUT_3V3, ISO_PPS_IN_3V3 are 3.3V level standards</p> <p>2, ISO_SYNC_IN, differential ISO_SYNC_IN (ISO_SYNC_IN_A, ISO_SYNC_IN_B) Maximum voltage 12V, low level (0~1V), high level (1~12V)</p> <p>3. ISO_SYNC_IN_A and ISO_SYNC_IN_B pins are suspended, ISO_SYNC_IN pin status is valid, ISO_SYNC_IN is pulled up to a high level, and differential signals (ISO_SYNC_IN_A and ISO_SYNC_IN_B) are valid</p> <p>4. See Appendix 3 for test methods.</p> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## 4.12 Multifunctional connector 2

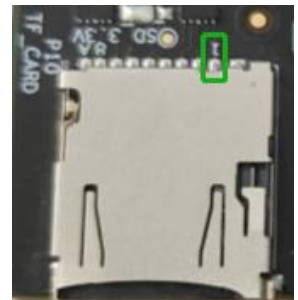
|            |                                       |             |     |             |
|------------|---------------------------------------|-------------|-----|-------------|
| Function   | Multifunctional connector             |             |     |             |
| Marking    | J8                                    |             |     |             |
| Type       | Multifunctional connector/ MX34040NF2 |             |     |             |
| Pin define | Pin                                   | Signal      | Pin | Signal      |
|            | 1                                     | GND         | 21  | GND         |
|            | 2                                     | RS232_1_TX  | 22  | RS232_2_TX  |
|            | 3                                     | RS232_1_RX  | 23  | RS232_2_RX  |
|            | 4                                     | RS232_1_RTS | 24  | RS232_2_RTS |
|            | 5                                     | RS232_1_CTS | 25  | RS232_2_CTS |
|            | 6                                     | GND         | 26  | GND         |
|            | 7                                     | IN1+        | 27  | OUT1+       |
|            | 8                                     | IN1-        | 28  | OUT1-       |
|            | 9                                     | IN2+        | 29  | OUT2+       |
|            | 10                                    | IN2-        | 30  | OUT2-       |
|            | 11                                    | IN3+        | 31  | OUT3+       |
|            | 12                                    | IN3-        | 32  | OUT3-       |
|            | 13                                    | IN4+        | 33  | OUT4+       |
|            | 14                                    | IN4-        | 34  | OUT4-       |
|            | 15                                    | GND         | 35  | GND         |
|            | 16                                    | CAN2.0_0_H  | 36  | CAN2.0_1_H  |
|            | 17                                    | CAN2.0_0_L  | 37  | CAN2.0_1_L  |
|            | 18                                    | GND         | 38  | GND         |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |           |    |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------|----|-----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 19 | RS485_2_B | 39 | RS485_1_B |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 20 | RS485_2_A | 40 | RS485_1_A |
| <p>Pin1: green box marked on the right.</p> <p>Pin2: red box marked on the right.</p> <p>In Linux, the mapping file for RS232_1 is ttyTHS0 in the /dev directory</p> <p>In Linux, the mapping file for RS232_2 is ttyTHS4 in the /dev directory</p> <p>In Linux, the mapping file for RS485_1 is ttyTHS3 in the /dev directory</p> <p>In Linux, the mapping file for RS485_2 is ttyTHS1 in the /dev directory</p> <p>In Linux,the mapping file for CAN_0 is can0 in the /dev directory</p> <p>In Linux,the mapping file for CAN_1 is can1 in the /dev directory</p> <p>The gpio number of IN1 in Linux is GPIO-341</p> <p>The gpio number of IN2 in Linux is GPIO-457</p> <p>The gpio number of IN3 in Linux is GPIO-452</p> <p>The gpio number of IN4 in Linux is GPIO-428</p> <p>The gpio number of OUT1 in Linux is GPIO-348</p> <p>The gpio number of OUT2 in Linux is GPIO-349</p> <p>The gpio number of OUT3 in Linux is GPIO-431</p> <p>The gpio number of OUT4 in Linux is GPIO-351</p> <p><b>GPIO and CAN driver loading methods: See Appendix 2</b></p> <p>Note:</p> <p>IN+ and IN- Voltage ranges from 5 to 48VDC</p> <p>OUT+, OUT- Maximum driving current 100mA</p> <p>The hardware of the RS485 supports switchover between incoming and outgoing packets, but the software does not need to control the switchover</p> |    |           |    |           |

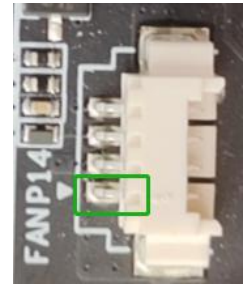
### 4.13 TF card holder

|                   |                                      |               |            |               |
|-------------------|--------------------------------------|---------------|------------|---------------|
| <b>Function</b>   | TF card holder                       |               |            |               |
| <b>Marking</b>    | P10                                  |               |            |               |
| <b>Type</b>       | TF card holder/TF-013                |               |            |               |
| <b>Pin define</b> | <b>Pin</b>                           | <b>Signal</b> | <b>Pin</b> | <b>Signal</b> |
|                   | 1                                    | DATA2         | 2          | DATA3/CD      |
|                   | 3                                    | CMD           | 4          | VDD           |
|                   | 5                                    | CLK           | 6          | VSS           |
|                   | 7                                    | DATA0         | 8          | DATA1         |
|                   | A                                    | DETECT        | B          | DETECT        |
|                   | Pin1: green box marked on the right. |               |            |               |



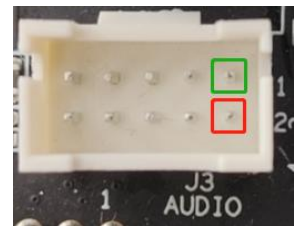
#### 4.14 Fan interface

| Function                                      | Connect the cooling fan                                                                                                                                                                       |        |     |         |     |        |     |        |   |     |   |         |   |      |   |     |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|---------|-----|--------|-----|--------|---|-----|---|---------|---|------|---|-----|
| Marking                                       | P14                                                                                                                                                                                           |        |     |         |     |        |     |        |   |     |   |         |   |      |   |     |
| Type                                          | 53261-0471                                                                                                                                                                                    |        |     |         |     |        |     |        |   |     |   |         |   |      |   |     |
| Pin define                                    | <table><tr><th>Pin</th><th>Signal</th><th>Pin</th><th>Signal</th></tr><tr><td>1</td><td>GND</td><td>2</td><td>FAN_VDD</td></tr><tr><td>3</td><td>TACH</td><td>4</td><td>PWM</td></tr></table> |        |     |         | Pin | Signal | Pin | Signal | 1 | GND | 2 | FAN_VDD | 3 | TACH | 4 | PWM |
|                                               | Pin                                                                                                                                                                                           | Signal | Pin | Signal  |     |        |     |        |   |     |   |         |   |      |   |     |
|                                               | 1                                                                                                                                                                                             | GND    | 2   | FAN_VDD |     |        |     |        |   |     |   |         |   |      |   |     |
|                                               | 3                                                                                                                                                                                             | TACH   | 4   | PWM     |     |        |     |        |   |     |   |         |   |      |   |     |
|                                               | Pin1: green box marked on the right.                                                                                                                                                          |        |     |         |     |        |     |        |   |     |   |         |   |      |   |     |
| Note: The default fan is 5V, 12V is optional. |                                                                                                                                                                                               |        |     |         |     |        |     |        |   |     |   |         |   |      |   |     |



#### 4.15 Audio interface

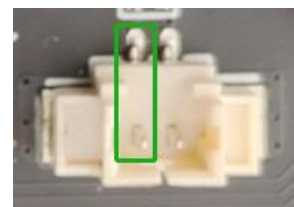
| Function                                                                                         | Connect audio devices                                                                                                                                                                                                                                                                                                                                                                                  |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|-----------|-----|--------|-----|--------|---|----------|---|---------|---|----------|---|---------|---|------|---|-----------|---|--------|---|-----------|---|-----------|----|-----------|
| Marking                                                                                          | J3                                                                                                                                                                                                                                                                                                                                                                                                     |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
| Type                                                                                             | PHB-2X5A                                                                                                                                                                                                                                                                                                                                                                                               |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
| Pin define                                                                                       | <table><tr><th>Pin</th><th>Signal</th><th>Pin</th><th>Signal</th></tr><tr><td>1</td><td>LINEIN_R</td><td>2</td><td>HPOUT_R</td></tr><tr><td>3</td><td>LINEIN_L</td><td>4</td><td>HPOUT_L</td></tr><tr><td>5</td><td>NULL</td><td>6</td><td>LINEOUT_R</td></tr><tr><td>7</td><td>MIC_IN</td><td>8</td><td>LINEOUT_L</td></tr><tr><td>9</td><td>AUDIO_GND</td><td>10</td><td>AUDIO_GND</td></tr></table> |           |     |           | Pin | Signal | Pin | Signal | 1 | LINEIN_R | 2 | HPOUT_R | 3 | LINEIN_L | 4 | HPOUT_L | 5 | NULL | 6 | LINEOUT_R | 7 | MIC_IN | 8 | LINEOUT_L | 9 | AUDIO_GND | 10 | AUDIO_GND |
|                                                                                                  | Pin                                                                                                                                                                                                                                                                                                                                                                                                    | Signal    | Pin | Signal    |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                      | LINEIN_R  | 2   | HPOUT_R   |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | 3                                                                                                                                                                                                                                                                                                                                                                                                      | LINEIN_L  | 4   | HPOUT_L   |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | 5                                                                                                                                                                                                                                                                                                                                                                                                      | NULL      | 6   | LINEOUT_R |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | 7                                                                                                                                                                                                                                                                                                                                                                                                      | MIC_IN    | 8   | LINEOUT_L |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | 9                                                                                                                                                                                                                                                                                                                                                                                                      | AUDIO_GND | 10  | AUDIO_GND |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  | Pin1: green box marked on the right.                                                                                                                                                                                                                                                                                                                                                                   |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
| Pin2: red box marked on the right.                                                               |                                                                                                                                                                                                                                                                                                                                                                                                        |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
| Note: In the /etc/realtimes directory, run the audio script./audio, see directory 4 for details. |                                                                                                                                                                                                                                                                                                                                                                                                        |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                        |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |
|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                        |           |     |           |     |        |     |        |   |          |   |         |   |          |   |         |   |      |   |           |   |        |   |           |   |           |    |           |



#### 4.16 RTC battery port

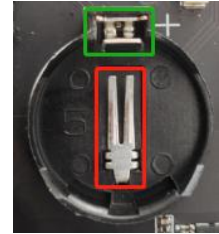
|            |                                      |        |     |        |
|------------|--------------------------------------|--------|-----|--------|
| Function   | RTC battery port                     |        |     |        |
| Marking    | P19                                  |        |     |        |
| Type       | AFC12-S02CCC-00                      |        |     |        |
| Pin define |                                      |        |     |        |
|            | Pin                                  | Signal | Pin | Signal |
|            | 1                                    | +      | 2   | -      |
|            | Pin1: green box marked on the right. |        |     |        |





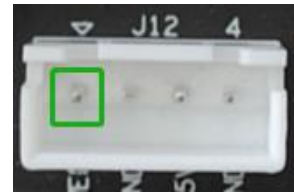
#### 4.17 RTC battery holder

|            |                                      |        |     |        |
|------------|--------------------------------------|--------|-----|--------|
| Function   | RTC battery holder                   |        |     |        |
| Marking    | BT1                                  |        |     |        |
| Type       | CR2032                               |        |     |        |
| Pin define |                                      |        |     |        |
|            | Pin                                  | Signal | Pin | Signal |
|            | 1                                    | +      | 2   | -      |
|            | Pin1: green box marked on the right. |        |     |        |
|            | Pin2: red box marked on the right.   |        |     |        |



#### 4.18 Expand the port for Power button

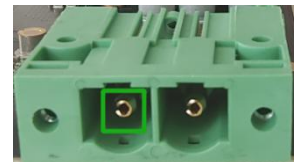
|                                                   |                                      |           |     |        |
|---------------------------------------------------|--------------------------------------|-----------|-----|--------|
| Function                                          | Expand the port for Power button     |           |     |        |
| Marking                                           | J12                                  |           |     |        |
| Type                                              | XH-2.54-4                            |           |     |        |
| Pin define                                        |                                      |           |     |        |
|                                                   | Pin                                  | Signal    | Pin | Signal |
|                                                   | 1                                    | POWER_BTN | 2   | GND    |
|                                                   | 3                                    | POWER_LED | 4   | GND    |
|                                                   | Pin1: green box marked on the right. |           |     |        |
| Note: POWER LED output voltage 5VDC, current 5mA. |                                      |           |     |        |



#### 4.19 Power input interface

|                                     |                                      |        |     |        |
|-------------------------------------|--------------------------------------|--------|-----|--------|
| Function                            | Power input interface                |        |     |        |
| Marking                             | J14                                  |        |     |        |
| Type                                | KF2EDGSRAM-10.16-2p                  |        |     |        |
| Pin define                          |                                      |        |     |        |
|                                     | Pin                                  | Signal | Pin | Signal |
|                                     | 1                                    | +      | 2   | -      |
|                                     | Pin1: green box marked on the right. |        |     |        |
| Power supply voltage range: +9~+36V |                                      |        |     |        |





#### 4.20 Power output interface

|                                      |                        |         |     |        |
|--------------------------------------|------------------------|---------|-----|--------|
| Function                             | Power output interface |         |     |        |
| Marking                              | P20                    |         |     |        |
| Type                                 | 1727010                |         |     |        |
| Pin define                           |                        |         |     |        |
|                                      | Pin                    | Signal  | Pin | Signal |
|                                      | 1                      | VCC_OUT | 2   | GND    |
| Pin1: green box marked on the right. |                        |         |     |        |



|  |                                                                                           |  |
|--|-------------------------------------------------------------------------------------------|--|
|  | VCC_OUT:9~36V consistent with the power supply voltage, the maximum withstand current 6A. |  |
|--|-------------------------------------------------------------------------------------------|--|

## 4.21 GMSL2 Expansion board Camera port

| <b>Function</b>   | GMSL2 Camera port                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |        |     |        |     |        |   |         |   |     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|--------|-----|--------|---|---------|---|-----|
| <b>Marking</b>    | J3~J10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |        |     |        |     |        |   |         |   |     |
| <b>Type</b>       | 7.03A0.00082D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |        |     |        |     |        |   |         |   |     |
| <b>Pin define</b> | <table border="1"> <thead> <tr> <th>Pin</th><th>Signal</th><th>Pin</th><th>Signal</th></tr> </thead> <tbody> <tr> <td>1</td><td>GMSL-IN</td><td>2</td><td>GND</td></tr> </tbody> </table> <p>Pin1: red box marked on the right.</p> <p><b>Note:</b></p> <p>J6-J3 connects to a camera device. The mapping files in the Linux operating system are /dev/video0-video3.</p> <p>J10-J7 connects to a camera device. The mapping files in the Linux operating system are /dev/video4-video7.</p> <p>Execute the script /etc/realtimes./ test_camera.sh to turn on the camera,</p> <p>If the screen flashes, switch to the root mode and run the./fsync-test file.</p> |     |        | Pin | Signal | Pin | Signal | 1 | GMSL-IN | 2 | GND |
| Pin               | Signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Pin | Signal |     |        |     |        |   |         |   |     |
| 1                 | GMSL-IN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2   | GND    |     |        |     |        |   |         |   |     |



## 4.22 SATA3.0 expansion interface

| <b>Function</b>   | 22Pin SATA hard drive socket                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|--------|-----|--------|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|------|---|------|----|------|----|-----|----|-----|----|-----|----|----|----|----|----|----|----|-----|----|------|----|-----|----|-----|----|-----|----|-----|
| <b>Marking</b>    | J3、J4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| <b>Type</b>       | LD2122F-S04T3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |        |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| <b>Pin define</b> | <table border="1"> <thead> <tr> <th>Pin</th><th>Signal</th><th>Pin</th><th>Signal</th></tr> </thead> <tbody> <tr> <td>1</td><td>GND</td><td>2</td><td>TX+</td></tr> <tr> <td>3</td><td>TX-</td><td>4</td><td>GND</td></tr> <tr> <td>5</td><td>RX+</td><td>6</td><td>RX-</td></tr> <tr> <td>7</td><td>GND</td><td>8</td><td>NULL</td></tr> <tr> <td>9</td><td>NULL</td><td>10</td><td>NULL</td></tr> <tr> <td>11</td><td>GND</td><td>12</td><td>GND</td></tr> <tr> <td>13</td><td>GND</td><td>14</td><td>5V</td></tr> <tr> <td>15</td><td>5V</td><td>16</td><td>5V</td></tr> <tr> <td>17</td><td>GND</td><td>18</td><td>NULL</td></tr> <tr> <td>19</td><td>GND</td><td>20</td><td>12V</td></tr> <tr> <td>21</td><td>12V</td><td>22</td><td>12V</td></tr> </tbody> </table> <p>Pin1: red box marked on the right.</p> |     |        | Pin | Signal | Pin | Signal | 1 | GND | 2 | TX+ | 3 | TX- | 4 | GND | 5 | RX+ | 6 | RX- | 7 | GND | 8 | NULL | 9 | NULL | 10 | NULL | 11 | GND | 12 | GND | 13 | GND | 14 | 5V | 15 | 5V | 16 | 5V | 17 | GND | 18 | NULL | 19 | GND | 20 | 12V | 21 | 12V | 22 | 12V |
| Pin               | Signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Pin | Signal |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 1                 | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2   | TX+    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 3                 | TX-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4   | GND    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 5                 | RX+                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6   | RX-    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 7                 | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8   | NULL   |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 9                 | NULL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10  | NULL   |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 11                | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 12  | GND    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 13                | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 14  | 5V     |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 15                | 5V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 16  | 5V     |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 17                | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 18  | NULL   |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 19                | GND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 20  | 12V    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |
| 21                | 12V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 22  | 12V    |     |        |     |        |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |      |   |      |    |      |    |     |    |     |    |     |    |    |    |    |    |    |    |     |    |      |    |     |    |     |    |     |    |     |



**Appendix 1:**

4G、5G network switcher:

The default port is M.2 key B (5G network). The switching modes between 4G and 5G networks are as follows:

1. Go to the /etc/realtimes directory and run the ./simcard\_switch.sh minipcie script to switch the SIM to the Mini PCIe port (4G network), power off, and restart.
2. Go to the /etc/realtimes directory and run the ./simcard\_switch.sh m2keyb script to switch the SIM to the M.2 Key B port (5G network), power off, and restart.

**Appendix 2:**

Usage of GPIO: Switch to the root mode

```
echo 348 > /sys/class/gpio/export //OUT1
echo 349 > /sys/class/gpio/export //OUT2
echo 431 > /sys/class/gpio/export //OUT3
echo 351 > /sys/class/gpio/export //OUT4
```

```
echo 1 or 0 > /sys/class/gpio/PA.00/value
echo 1 or 0 > /sys/class/gpio/PA.01/value
echo 1 or 0 > /sys/class/gpio/PM.07/value
echo 1 or 0 > /sys/class/gpio/PA.03/value
```

```
echo 341 > /sys/class/gpio/export //IN1
echo 457 > /sys/class/gpio/export //IN2
echo 452 > /sys/class/gpio/export //IN3
echo 428 > /sys/class/gpio/export //IN4
```

```
cat /sys/class/gpio/PEE.02/value
cat /sys/class/gpio/PR.01/value
cat /sys/class/gpio/PQ.04/value
cat /sys/class/gpio/PM.04/value
```

CAN driver loading:

```
sudo modprobe can // Load the can module
sudo modprobe can-raw
sudo modprobe mttcan
sudo ip link set can0 up type can bitrate 500000 // Set the baud rate
sudo ip link set can1 up type can bitrate 500000 // Set the baud rate
candump can0 & // Receive can0 data
candump can1 & // Receive can1 data
cansend can0 123#abcd // Send can0 data
cansend can1 123#abcd // Send can1 data
```

### Appendix 3:

Sync-in/out test method:

1. Short-circuit Sync out to Sync in.
2. Initiate the timestamp event by entering the following command to start the application:

```
./gte_mon -d gtechip0 -g 324 -r -f
```

GPIO324 is mapped to port BB0 of AON, that is, SYNC-IN.

Data input detected:

```
nvidia@tegra-ubuntu:~$ sudo ./gte_mon -d gtechip0 -g 324 -r -f
[sudo] password for nvidia:
Monitoring line 324 on gtechip0
HW timestamp GPIO EVENT 2493772432832
HW timestamp GPIO EVENT 2493872466720
HW timestamp GPIO EVENT 2493972432800
HW timestamp GPIO EVENT 2494072466688
HW timestamp GPIO EVENT 2494172432832
HW timestamp GPIO EVENT 2494272466688
HW timestamp GPIO EVENT 2494272466688
HW timestamp GPIO EVENT 2494272466688
HW timestamp GPIO EVENT 2494372432832
HW timestamp GPIO EVENT 2494472466688
HW timestamp GPIO EVENT 2494572432832
HW timestamp GPIO EVENT 2494672466752
HW timestamp GPIO EVENT 2494772432832
HW timestamp GPIO EVENT 2494872466688
HW timestamp GPIO EVENT 2494972432800
```

Sync-pps test method:

1. Whenever the GPS PPS comes, an interrupt signal will be generated at the corresponding GPIO falling edge, and a timestamp will be generated at this time, which can be viewed by the following command:

```
n@n-desktop:~$ cat /sys/class/pps/pps0/assert
1691986383.752656412#3521
```

Test PPS:

Installation tools: sudo apt install pps-tools

Use the command sudo pptest /dev/pps0 to see output similar to the following:

```
trying PPS source "/dev/pps0"
found PPS source "/dev/pps0"
ok, found 1 source(s), now start fetching data...
source 0 - assert 1421066632.974111422, sequence: 1698 - clear 0.000000000, sequence: 0
source 0 - assert 1421066633.974045488, sequence: 1699 - clear 0.000000000, sequence: 0
```

Run the cat /proc/interrupts | grep pps command to check the number of interrupts of the PPS:

1. GPSD installation:

GPSD is a special software used to obtain and parse GPS data, which works in the form of a daemon process, and places the GPS data after obtaining and parsing on a specific TCP port for further reading by other applications (such as: ntpd chronyd).

To install third-party tools:

```
sudo apt-get install -y setserial gpsd gpsd-clients python-gps pps-tools
```

Modify the gpsd file in the /etc/default directory:

```
+DEVICES="/dev/ttyTHS1" //GPS sensor System port number
```

```
+GPSD_OPTIONS="-n -G"
```

Annotation:

-n = don't wait for client connects to poll GPS

-G = make gpsd listen on INADDR\_ANY

```
sudo stty -F /dev/ttyTHS1 38400
```

```
sudo gpsd /dev/ttyTHS1 -F /var/run/gpsd.sock
```

gpsd running service (starting from boot) :

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable gpsd
```

```
sudo systemctl restart gpsd
```

At this time, you can use gpsmon or cgps tools to display GPS data parameters in a graphical interface.

**Note: If the gpsmon GUI fails to open, run the `sudo stty -F /dev/ttyTHS1 38400` command again.**

## 2. chrony installation:

chrony is a dedicated time server program that reads the data analyzed by GPSD. GPSD synchronizes system time by obtaining the time service provided by GPSD, similar to NTPD.

Installation command:

```
sudo apt install chrony -y
```

Then modify the /etc/chrony/chrony.conf file.

```
# Log files location.
logdir /var/log/chrony

maxdistance 16.0

# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0

# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
leapsectz right/UTC
#makestep 1.0 -1
rtcsync

refclock PPS /dev/pps0 lock GPSD prefer refid PPS
refclock SHM 0 offset 0.0 delay 0.3 refid GPSD

allow

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3
```

rtcsync --- The instruction will enable a kernel mode in which system time is copied to the Real-Time Clock (RTC) every 11 minutes.



allow / deny --- Here you can specify a host, subnet, or network to allow or deny NTP connections to machines acting as clock servers.

makestep --- Typically, chronyd will allow the system to gradually correct all time deviations by slowing down or speeding up the clock as needed. In some cases, the system clock may drift too fast, causing the adjustment process to take a long time to correct the system clock. This directive forces chronyd to step the system clock when the adjustment period is above a certain threshold, but only takes effect if there are no more clock updates because chronyd boot time exceeds the specified limit (a negative value can be used to disable the limit).

After setting parameters, restart the chrony service:

```
sudo service chrony restart
```

After the above preparation is complete, the test time synchronization.

View the time synchronization source and status:

```
chronyc sources
```

```
chronyc sources -v
```

```
chronyc sourcestats -v
```

```
systemctl status chronyd // View the service status
```

```
chronyc tracking // Calibrate the time server
```

```
timedatectl
```

For example:

```
z@z-desktop:/$ chronyc sources
```

```
210 Number of sources = 2
```

```
MS Name/IP address          Stratum Poll Reach LastRx Last sample
```

```
=====  
#? PPS                      0  4    0  771  -202ms[ -6056h] +/-  39us  
#* GPSD                     0  4  377   16   -40ms[  -43ms] +/- 167ms
```

Note: # preceded by a question mark "?" Or x, indicating that the synchronization is not successful.

Asterisk \*, indicating that the synchronization is successful.

```
chronyc sources -v
```



chronyc sourcestats -v

systemctl status chronyd // View the service status

- chrony.service - chrony, an NTP client/server

Loaded: loaded (/lib/systemd/system/chrony.service; enabled; vendor preset: e

Active: active (running) since Wed 2021-07-21 14:51:40 EDT; 8 months 8 days a

Docs: man:chronyd(8)

man:chronyc(1)

man:chrony.conf(5)

Main PID: 5527 (chronyd)

Tasks: 1 (limit: 4915)

CGroup: /system.slice/chrony.service

└─5527 /usr/sbin/chronyd

Jul 21 14:51:40 z-desktop systemd[1]: Starting chrony, an NTP client/server...

Jul 21 14:51:40 z-desktop chronyd[5527]: chronyd version 3.2 starting (+CMDMON +

Jul 21 14:51:40 z-desktop chronyd[5527]: Frequency -5.074 +/- 8.364 ppm read fro

Jul 21 14:51:40 z-desktop chronyd[5527]: Using right/UTC timezone to obtain leap

Jul 21 14:51:40 z-desktop systemd[1]: Started chrony, an NTP client/server.

Jul 21 15:02:20 z-desktop chronyd[5527]: Selected source GPSD

Jul 21 15:02:20 z-desktop chronyd[5527]: System clock TAI offset set to 37 secon

Jul 21 15:02:20 z-desktop chronyd[5527]: System clock wrong by 21801086.045778 s

Mar 30 22:53:46 z-desktop chronyd[5527]: System clock was stepped by 21801086.04

chronyc tracking

Reference ID : 47505344 (GPSD)

Stratum : 1

Ref time (UTC) : Thu Mar 31 03:03:45 2022

System time : 0.001609250 seconds slow of NTP time

Last offset : -0.001323730 seconds

RMS offset : 0.001518739 seconds

Frequency : 35.957 ppm fast

Residual freq : -0.735 ppm

Skew : 27.275 ppm

Root delay : 0.300000012 seconds

Root dispersion : 0.018442914 seconds

Update interval : 16.0 seconds

Leap status : Normal

z@z-desktop:/\$ timedatectl

Local time: Wed 2022-03-30 23:28:32 EDT

Universal time: Thu 2022-03-31 03:28:32 UTC

RTC time: Thu 2022-03-31 03:28:32

Time zone: America/New\_York (EDT, -0400)

System clock synchronized: yes



systemd-timesyncd.service active: yes

RTC in local TZ: no

#### Appendix 4:

There are two types of audio input: line\_in and mic\_in.

In the /etc/realtime directory, edit the audio script./audio.sh, and use line\_in to modify the audio script as follows:

```
amixer -c 1 cset name="H40-SGTL Capture Mux" "LINE_IN"
```

```
#amixer -c 1 cset name="H40-SGTL Capture Mux" "MIC_IN"
```

After editing, save and exit and run./audio.sh

Audio input mic\_in mode Changes as follows:

```
#amixer -c 1 cset name="H40-SGTL Capture Mux" "LINE_IN"
```

```
amixer -c 1 cset name="H40-SGTL Capture Mux" "MIC_IN"
```

After editing, save and exit and run./audio.sh

#### Appendix 5:

Usage of GPIO: Switch to the root mode

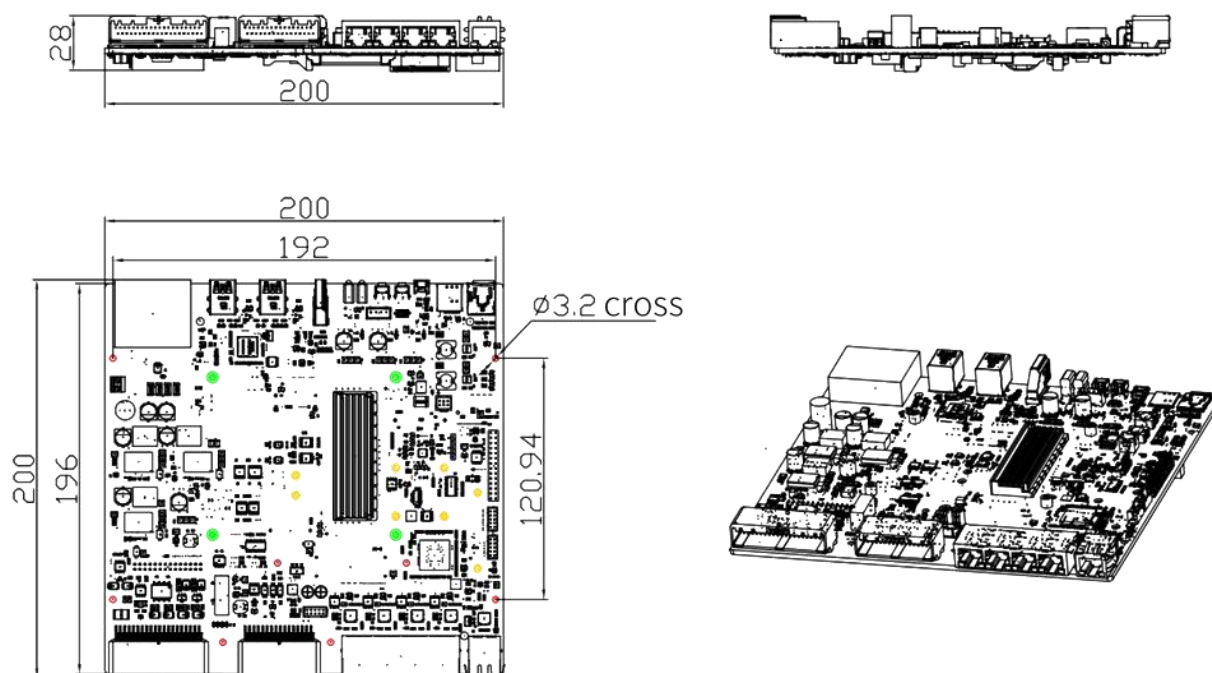
```
echo 456 > /sys/class/gpio/export //GPIO01
```

```
echo 485 > /sys/class/gpio/export //GPIO02
```

```
echo 1 or 0 > /sys/class/gpio/PR.01/value //GPIO01
```

```
echo 1 or 0 > /sys/class/gpio/PZ.07/value //GPIO02
```

## 5 Product size



## 6 Hardware update history

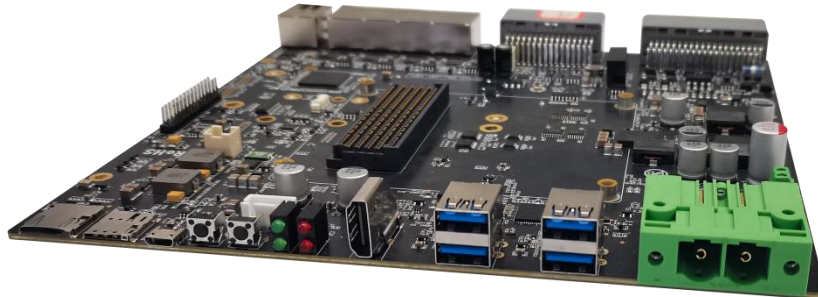
RTSO-2002 Board hardware update history

| Version | Update description |
|---------|--------------------|
| v1.0    | EVT version        |
| v2.0    | DVT version        |

# Software package description

## 1 Installation and use

### 1.1 Picture of products



### 1.2 How to use the board

Before using the RTSO-2002 carrier board, you must verify that the Jetson AGX Orin core module is installed with the operating system loaded with Realtimes BSP.

- a) Make sure all external system voltages are off
- b) Install the Jetson AGX Orin core module onto the 699-Pin SO-DIMM connector. Ensure that the connectors are aligned with even force while installing the fixing screws.
- c) Install necessary external cables. (e.g. Cable to HDMI display, power input cable to power system, USB cable to connect keyboard and mouse...)
- d) Connect the power cord to the power supply.
- e) RTSO-2002 adopts automatic power-on design, power on, and the system starts to work.
- f) For a system without a protective cover, do not move the entire system after the system is powered on. Do not touch the circuit board or its electronic components with your body.

### 1.3 Recovery mode

Jetson AGX Orin core module can work in normal mode and Recovery mode. In Recovery mode, file system update, kernel update, boot loader update, BCT update and other operations can be performed.

To enter the Recovery mode, perform the following steps:

- a) Power off the system.
- b) Use a Micro-USB cable to connect the Micro-USB port (P7) of the Jetson AGX Orin to the Host(PC) to develop the host USB port.
- c) Press the RECOVERY button to power on the system. Hold the power supply for more than 3 seconds, and then release the RECOVERY button.
- d) When the system enters Recovery mode, you can perform subsequent operations.

## 2 Board Support Package (BSP)

When the RTSO-2002 board works on a system burned using the official NVIDIA Linux For Tegra (L4T),



most of the hardware interfaces do not work properly, and the full support of the RTSO-2002 on-board interface requires a supporting driver patch.

NVIDIA's original LT4 package can be downloaded from the following link:

**<https://developer.nvidia.com/embedded/linux-tegra>**

For the RTSO-2002 driver patch support package, contact Realtimes.

## 3 Other related introductions

### 3.1 Jetpack

Nvidia jetpack sdk is the most comprehensive solution for building AI applications. It includes software libraries and APIs, samples, developer tools and documentation for the latest and previous versions of Jetson products.

#### 3.1.1 Jetpack Component summary

This section briefly introduces each component of JetPack. For more detailed information about these components, please refer to JetPack's online documentation.

##### OS Image

JetPack includes a reference file system derived from Ubuntu. (Development kit system, no need to install)

##### Libraries and APIs

JetPack library and API include:

- TensorRT and cuDNN for high-performance deep learning applications
- CUDA for multi-domain GPU accelerated applications
- NVIDIA Container Runtime for containerized GPU accelerated applications
- Multimedia API package for camera application and sensor driver development
- VisionWorks, OpenCV and VPI for visual computing applications
- Sample application

#### 3.1.2 How to install Jetpack

##### Overview of process steps

Installing JetPack to your Jetson device requires you to perform the following steps:

1. Download and install the NVIDIA SDK Manager on the Linux host.
2. Connect your jetson device to the Linux host (Micro USB or network (same network segment)).
3. Test your device can connect via ssh.
4. Use SDK Manager to select and install the required components.

##### Download and install the NVIDIA SDK Manager on the Linux host.

You must have a Linux host with internet access to run SDK Manager and refresh the developer kit. The supported host operating systems are:

Ubuntu Linux x64 Version 18.04 or 16.04



Download and install NVIDIA SDK Manager.

**sdkmanager** Installation package: **sdkmanager-[version].[build#].deb**

The current version is: **sdkmanager\_1.2.0**

**sdkmanager** install

**\$ sudo apt install ./sdkmanager-[version].[build#].deb**

### **Connect your jetson device to the Linux host**

Prepare your Jetson device for the following settings. (If the equipment system is not initialized, the system user needs to initialize after power-on)

- Connect the monitor, keyboard and mouse to the Jetson Feiyun smart box device (please refer to the interface description above).
- Use a USB cable to connect the Linux host and the Micro USB OTG interface of the RTSO-3004 carrier board (or network (same network segment))
- During the setup process, SDK Manager will provide internet connection to your jetson device via Micro USB or network connection.
- Connect the supplied power adapter to the DC jack of the Jetson device and plug the AC port into an AC outlet.

### **Test that your device can connect via ssh.**

If connected via Micro USB, enter lsusb under the Linux host terminal to see the nvidia crop

The device can be accessed through ping or ssh commands

**\$ ping 192.168.55.1**

or

**\$ ssh <jetson\_user>@192.168.55.1**

If connected via a network, check whether the device ip address is in the same network segment as the Linux host on the Feiyun Smart Box, and the Feiyun Smart Box device can be accessed through the ping or ssh command.

**\$ ping <jetson\_ip>**

or

**\$ ssh <jetson\_user>@<jetson\_device\_ip>**

### **Use SDK Manager to select and install the required components**

NVIDIA SDK Manager supports installing software to Jetson core (Flying Cloud Smart Box). For complete instructions, please refer to Chapter 5 of "[Orin NX System Programming Manual](#)".

## **3.2 Developer tools**

JetPack includes the following development tools. Some are used directly on the Jetson system, and some run on a Linux host connected to the Jetson system.



- **Application development and debugging tools**

- NSight Eclipse version for GPU-accelerated application development: runs on a Linux host. All Jetson products are supported.
- CUDA-GDB for application debugging: Runs on a Jetson system or a Linux host. All Jetson products are supported.
- CUDA-MEMCHECK to debug application memory errors: run on Jetson systems. All Jetson products are supported.

- **Application analysis and optimization tools**

- NSight Systems for application multi-core CPU profiling: runs on a Linux host. Helps you improve application performance by identifying slow parts of your code. All Jetson products are supported.
- NVIDIA® Nsight™ compute kernel profiler: an interactive profiling tool for CUDA applications. It provides detailed performance metrics and API debugging through the user interface and command line tools.
- NSight Graphics for Graphics Application Debugging and Profiling: A consolegrade tool for debugging and optimizing OpenGL and OpenGL ES programs. Running on a Linux host. All Jetson products are supported.

### 3.3 Documentation

Documents related to developers using JetPack include:

- [JetPack Documentation](#)
- [VisionWorks Documentation](#)
- [Nsight Eclipse Edition Documentation](#)
- [CUDA-GDB Documentation](#)
- [CUDA-MEMCHECK Documentation](#)
- [TensorRT Documentation](#)
- [cuDNN Documentation](#)
- [CUDA Toolkit](#)
- [NVIDIA Container Runtime](#)
- [OpenCV Documentation](#)
- [Jetson Linux Multimedia API Reference](#)
- [Nsight Systems](#)
- [nvprof](#)
- [Visual Profiler](#)
- [Nsight Graphics](#)
- [Nsight Compute CLI](#)
- [VPI–Vision Programming Interface](#)



## Terms of Warranty

### Important note

Each embedded product provided by Realtimes Technology is free from any defects in material and process, fully in line with the specifications officially issued by the original factory.

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If the product does not work properly, Please contact Realtimes Technology or dealer for warranty service, please show invoice when product warranty (this is the proof for you getting warranty service).

### Warranty solution

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- c) Personal unauthorized changing the product (such as changing circuit characteristics, mechanical characteristics, software characteristics, Conformal coating).
- d) Other failures which are clearly due to misuse (such as overvoltage, polarity reversal, the pin bent or broken, the wrong connection, drop damage, transportation damage, damage due to over operating temperature and so on).
- e) The logo and part number on the product have been deleted or removed.
- f) The product is out of warranty.

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If the same fault multiple occurrence for the products,in order to find out the reason causing the problem,we will request the users to provide the specific documents or information of peripheral equipment,such as monitor,I/O boards,cables,power supply, diagram and structure of the system,etc.If such documents or information are not available,we have the right to refuse to perform the warranty, the repairments will be charged accordingly.

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